

Book the Doc

Modernizing Access to Doctor Appointments



Project Report

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1.Team Members

Group - 5

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2.Executive Summary

This report outlines the design, development, and deployment of "Book the Doc," a digital healthcare appointment and diagnostic test booking platform tailored for small and medium-sized medical centers in Sri Lanka. The platform empowers medical centers to manage doctor sessions and lab services, while patients can conveniently search, book, and monitor appointments and queue positions in real time.

Currently, Book the Doc supports secure appointment placement with on-site payments and real-time queue tracking. Built using modern technologies like Next.js, Tailwind CSS, Zustand, Node.js, and MongoDB Atlas, it ensures security, scalability, and a smooth user experience. Future features include telemedicine, AI-based diagnosis assistance, e-payment integration, and machine learning for appointment time predictions.

The project addresses key inefficiencies in Sri Lanka's healthcare system by offering a centralized, accessible, and user centric solution that reduces patient wait times and modernizes clinic operations.

3.Problem Statement

Access to timely and efficient healthcare services in Sri Lanka remains a challenge, particularly in medium and small-scale medical centers. Many of these institutions still rely on outdated manual booking systems, leading to long queues, miscommunication, and poor patient experiences. Traditional appointment methods such as walk-ins or phone calls are inefficient in today's fast-paced, digital-first environment.

Furthermore, while e-channeling platforms like Doc990 serve larger hospitals, they fail to support independent clinics and smaller facilities. The lack of doctor feedback, poor UI design, and absence of real-time queue tracking in such platforms limit their utility.

There is a critical need for a centralized, scalable, and intuitive digital solution that addresses these shortcomings by enabling online appointment bookings, test reservations, live queue updates, and future integrations like e-payments and AI-powered medical support.

4. Market Analysis

The current healthcare appointment and channeling systems, particularly in medium to small medical centers, exhibit several inefficiencies and technological gaps. A significant portion of these facilities still rely heavily on manual processes, resulting in prolonged waiting times and operational bottlenecks. Over Reliance on manual methods not only decreases efficiency but also leads to increased chances of human error and miscommunication.

Moreover, the lack of a centralized system across healthcare facilities has contributed to fragmented services. Patients often face challenges such as long queues, waiting times, and difficulty in accessing real-time information about doctor availability. This situation is worsened in smaller centers where the use of technology remains limited.

While some existing solutions like **Doc990** and **Asiri Hospitals** platforms provide digital access to healthcare services, they are primarily designed for larger hospitals. Medium and small centers often do not have their own dedicated platforms. Additionally, platforms like Doc990 lack user-centric features such as doctor ratings or feedback systems, making it difficult for patients to make informed decisions. Their interfaces are also noted for poor user experience, making it hard to navigate or find suitable doctors easily.

This market gap highlights a clear demand for a more accessible, user friendly, and centralized digital solution that caters to all sizes of medical centers, especially in underserved regions.

5. Feasibility Study

5.1 Financial Feasibility

Book The Doc utilizes a subscription based model that significantly reduces upfront costs for small to mid-sized medical centers, making it accessible without the heavy investment typical of enterprise healthcare solutions. By deploying on Vercel, using Railway for backend services, and MongoDB Atlas for database hosting, the platform leverages free-tier and scalable pricing plans, minimizing hosting and infrastructure expenses.

This cost effective technology stack avoids the need for on-premise servers or complex infrastructure. The integration of free tools like **Google OAuth** for authentication and **Nodemailer** for email sending further decreases operational costs, aligning the platform with the financial constraints of smaller clinics and independent practitioners.

5.2 Technical Feasibility

The solution is technically feasible and has already been deployed at <https://book-the-doc.vercel.app>. It uses:

- **Frontend:** Built with **Next.js** and **Tailwind CSS**, providing fast performance and modern UI design. Used **Zustand** for state management and **Formik** for frontend validation.
- **Backend:** Powered by **Node.js**, ensuring non-blocking I/O for real-time updates and concurrent booking requests.
- **Database:** Uses **MongoDB**, a scalable, NoSQL cloud database that is well-suited for dynamic data models like appointments and patient profiles.
- **Hosting:** **Vercel** and **Railway** offer reliable hosting.
- **Authentication:** Uses **Google OAuth 2.0** for google signIn.

The technology choices ensure cross-platform compatibility, quick load times, and easy maintenance. Smart features like AI chat-bots and queue prediction are technically supported by the selected stack.

5.3 Resource and Time Feasibility

5.3.1 Resource Feasibility:

The development team has already completed a functional prototype, proving that the necessary technical and human resources are in place. Technologies used are open-source or have generous free tiers, requiring no additional licenses.

5.3.1 Time Feasibility:

Given the project's prototype is already operational, expanding it into a fully functional product can be completed within a few months. Minor integration and user testing would be required before broader deployment. Risks related to academic or personal time constraints can be mitigated through agile development practices and weekly reviews.

5.4 Risk Feasibility

Handling sensitive medical data on a public platform poses significant security and privacy challenges. *Book The Doc* addresses these concerns through multiple layers of protection. The platform employs **Google OAuth 2.0** for secure user authentication, minimizing exposure to credential theft. Additionally, it implements **JWT (JSON Web Tokens)** for stateless session management, enabling scalable and secure handling of authenticated requests.

To further enhance security, a two token based authentication system is used, combining access and refresh tokens to maintain session integrity while reducing the risk of token abuse. All passwords and sensitive user data are securely hash using **bcrypt**, ensuring that even if data were compromised, credentials would remain protected.

Although the current deployment uses **Railway** for hosting, production environments will incorporate more robust risk mitigation strategies including encrypted database entries, secure HTTPS endpoints, and routine data backups. As the system evolves, plans include supporting encrypted electronic medical records and integration with governmental eHealth APIs, which will not only

bolster security but also enhance system interoperability and reliability across Sri Lanka's healthcare infrastructure.

5.5 Social/Legal Feasibility

5.5.1 Social Feasibility:

The platform enhances patient experience by reducing wait times, simplifying appointment booking, and increasing transparency through doctor profiles and patient feedback. Resistance may arise from medical staff unfamiliar with digital tools, which can be addressed through training and intuitive UI design.

5.5.2 Legal Feasibility:

The system must comply with **Sri Lanka's Personal Data Protection Act (PDPA)** and healthcare regulations. Though the current prototype doesn't include direct medical record storage, any future inclusion must ensure encrypted storage, consent mechanisms, and secure access control. Regular audits and compliance with healthcare IT standards are essential for legal viability.

6. Project Scope

The scope of this project encompasses the design, development, and deployment of a web-based healthcare appointment and diagnostic lab booking system that serves the needs of various user groups, including patients, medical centers, and administrators. The system is structured to provide tailored functionalities for each user role, ensuring a comprehensive, efficient, and scalable digital solution that modernizes healthcare access, streamlines medical operations, and improves patient satisfaction.

6.1 Patient Interface

Guest and Registered Users: The platform accommodates both guest and registered patients. Registered users benefit from features such as saved booking history, pre-filled appointment forms, access to upcoming appointment schedules, and personalized doctor or test recommendations based on previous interactions.

Appointment Booking and Live Queue Tracking: Patients can browse available doctors and lab tests, view doctor availability and session times, and book appointments with a few simple steps. After booking, patients can monitor their real-time position in the live queue, reducing unnecessary waiting at the center.

User Account Management: Registered patients have full control over their profile information, including personal details, upcoming appointments, and past booking history. They can also cancel or reschedule appointments when necessary.

6.2 Center (Medical Staff) Interface

Session and Lab Test Management: Medical centers can add doctor sessions with detailed availability schedules, as well as manage their diagnostic test offerings (e.g., blood tests, X-rays). The interface allows for creating, updating, and removing session and test information easily.

Queue Management and Real-time Updates: Staff can view and manage real-time patient queues. This includes marking patients as "arrived", "in consultation", or "completed", which helps keep the system updated and patients informed.

Request and Appointment Handling: Centers can manage appointment requests, cancellations, and custom notifications to patients. Optional features like setting daily limits, buffer times, and closing periods are also available.

6.3 Administrator Interface

Analytics Dashboard: Admin users have access to a centralized analytics dashboard that tracks platform usage, popular appointment times, common test types, and patient behavior. These insights aid in platform performance monitoring and decision-making.

Platform Oversight: Administrators oversee the onboarding of medical centers, manage reported issues, and ensure compliance with data protection policies. They can assign or revoke access permissions for centers and manage system-wide settings.

Content and Feature Management: Admins can publish announcements, update FAQs, and push updates or features to either patient or center portals.

6.4 System Integration and Compliance

Scalability and Mobile App Preparation: The system is designed for scalability and future enhancements. A mobile application is planned for development to allow patients to book appointments and track queues conveniently on mobile, while centers can update queues on the go.

Data Security and Privacy: Patient data security is ensured using JWT-based two-token authentication, bcrypt password hashing, and HTTPS-secured endpoints. The system complies with local healthcare data policies and is built to scale toward international standards such as GDPR.

6.5 Support and Maintenance

Continuous Monitoring and Support: Post-deployment, the system includes a dedicated support channel and automated monitoring tools to ensure 24/7 availability. Error tracking and usage logs are used for rapid debugging and user assistance.

Upgrades and Feature Expansion: The architecture supports regular feature rollouts, including AI-based doctor suggestions, symptom-based chatbot triaging, and integration with e-payment gateways. The platform can be upgraded modularly without affecting core operations.

7. Business Model Canvas

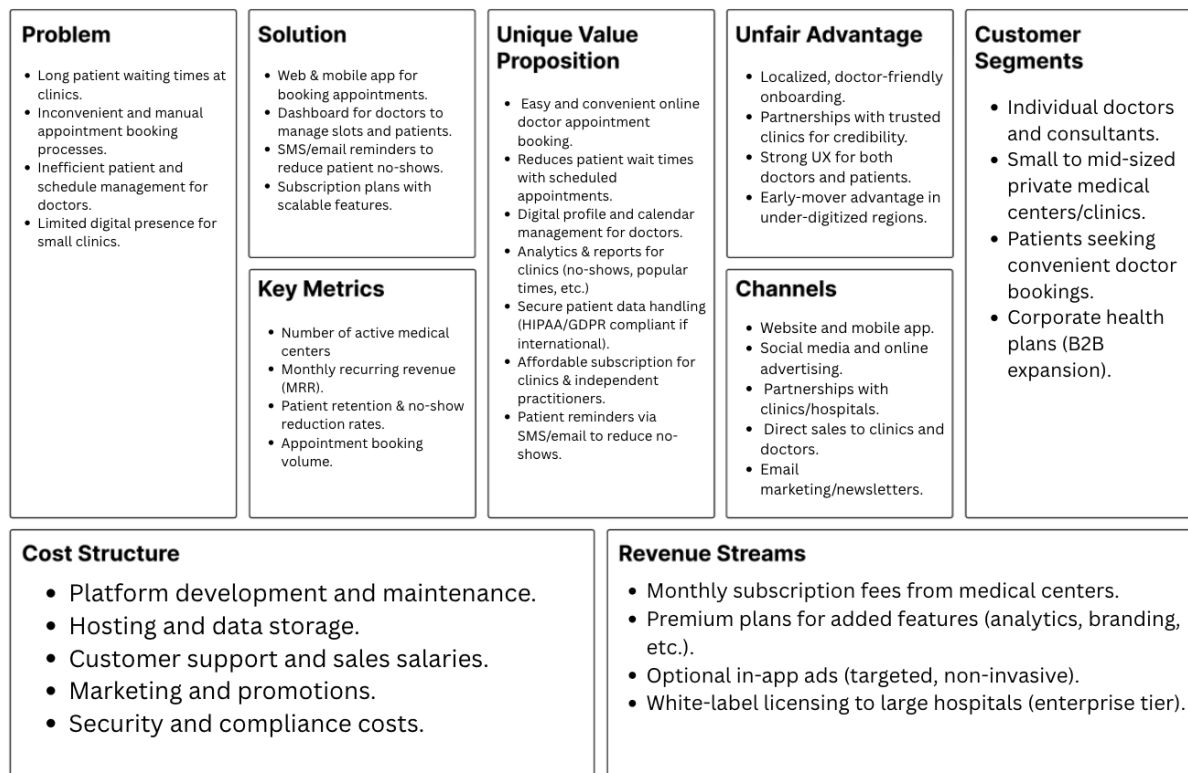


Figure 1 - Business Model Canvas

8. Unique Value Proposition

- Easy and convenient online doctor appointment booking.
- Reduces patient wait times with scheduled appointments.
- Digital profile and calendar management for doctors.
- Analytics & reports for clinics (no-shows, popular times, etc.)
- Secure patient data handling
- Affordable subscription for clinics & independent practitioners.
- Patient reminders via SMS/email to reduce no-shows.

9. Proposed Solution (Architecture and Features)

Our platform simplifies scheduling doctor appointments through an easy to use web interface and mobile application. Doctors can register their practices, enhancing their visibility and reaching potential patients. Users can find nearby doctors and book appointments online, receiving a unique channel number for real time queue tracking and updates. Our system uses machine learning to predict wait times, enabling users to plan their schedules effectively. Additionally, we recommend nearby centers based on illness type and other details.

This solution streamlines the booking process, benefiting both patients and healthcare providers. Patients enjoy a convenient experience, saving time and reducing frustration, while doctors can manage patient queues more efficiently, improving overall healthcare delivery and creating a more organized environment.

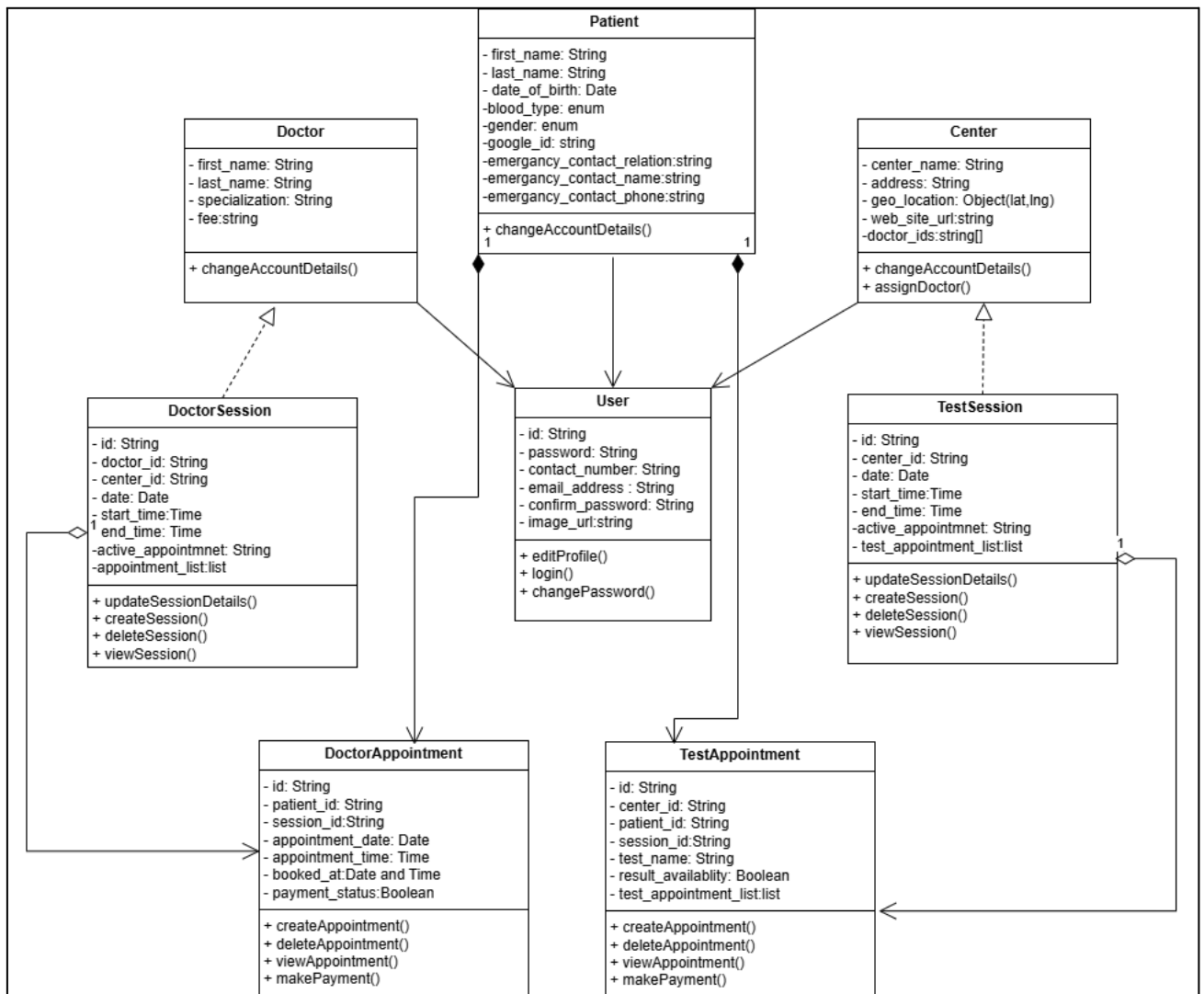


Figure 2 - Class Diagram

10. Implementation

10.1 Main Interfaces

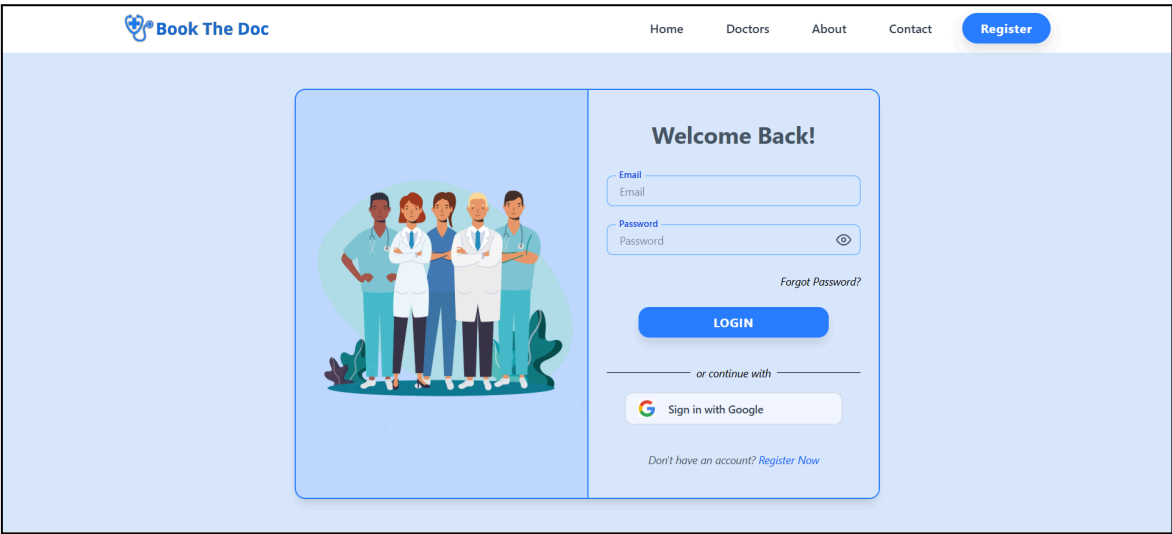


Figure 3 - Login Screen

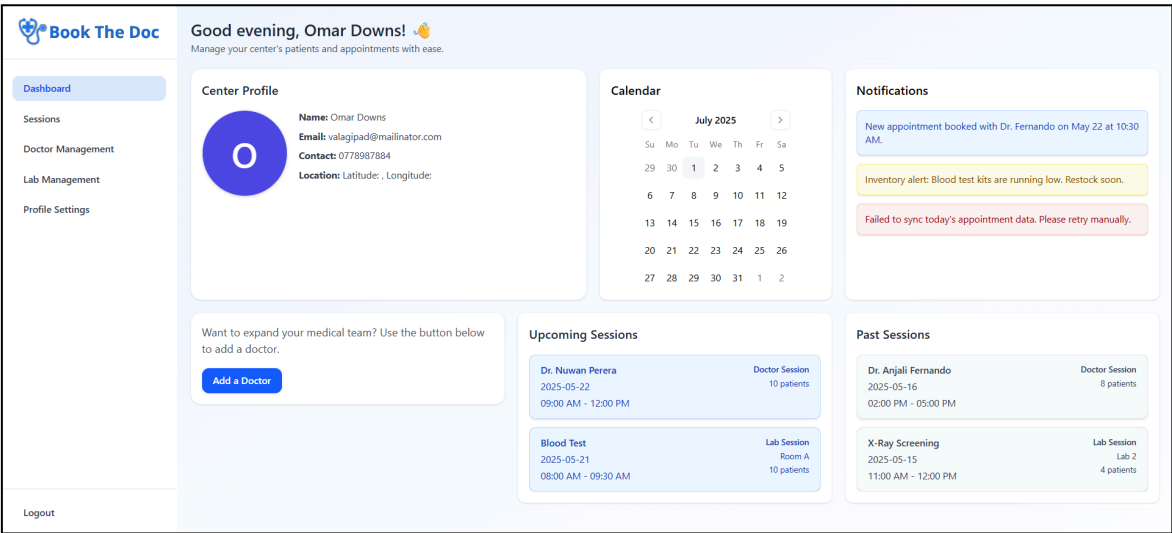


Figure 4 - Center Dashboard Screen

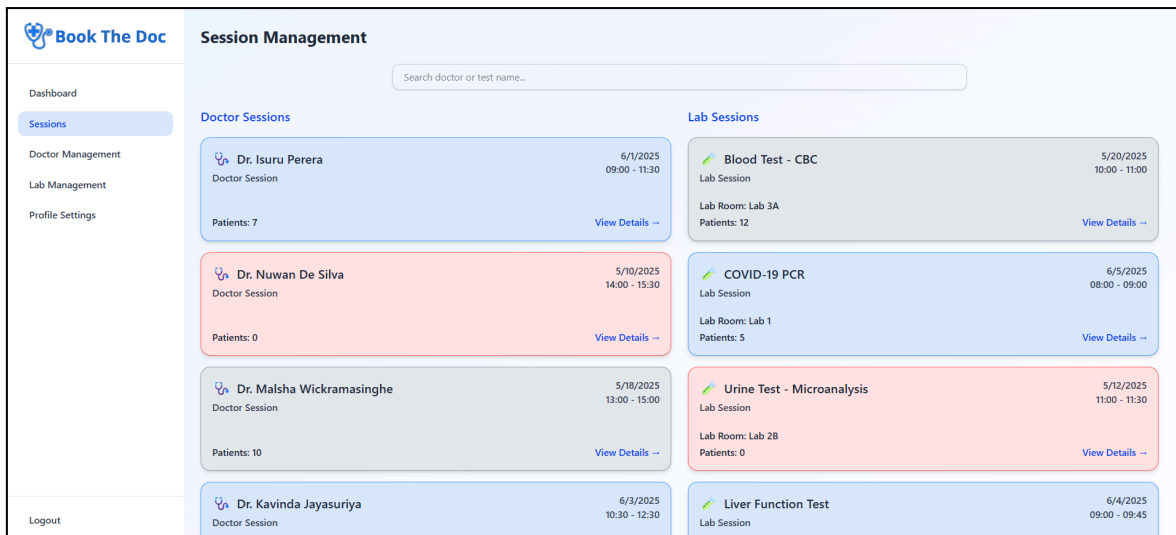


Figure 5 - Session Management Screen

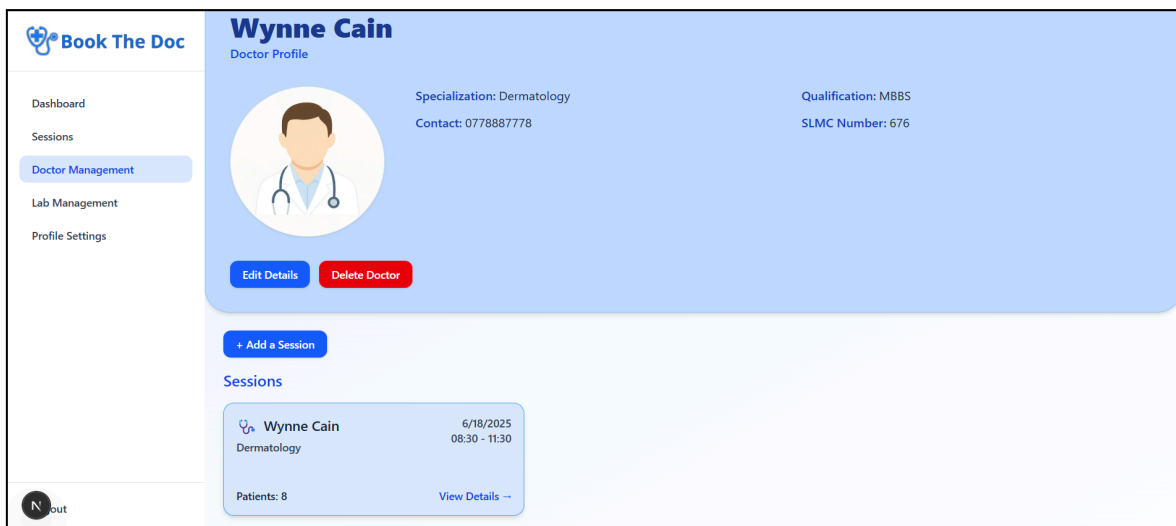


Figure 6 - Doctor Session Screen

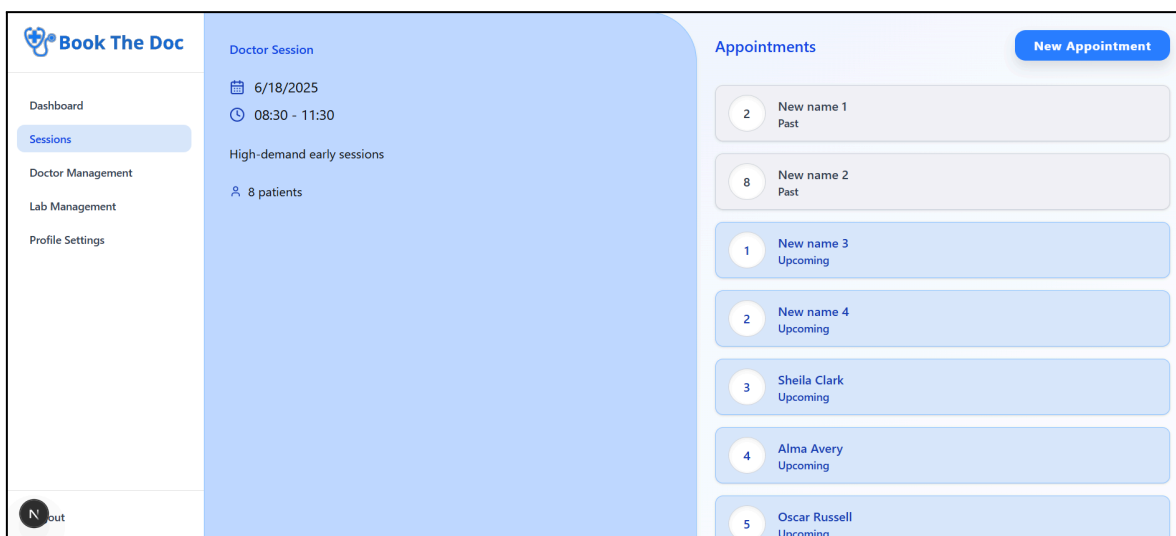


Figure 7 - Patient Appointments Screen

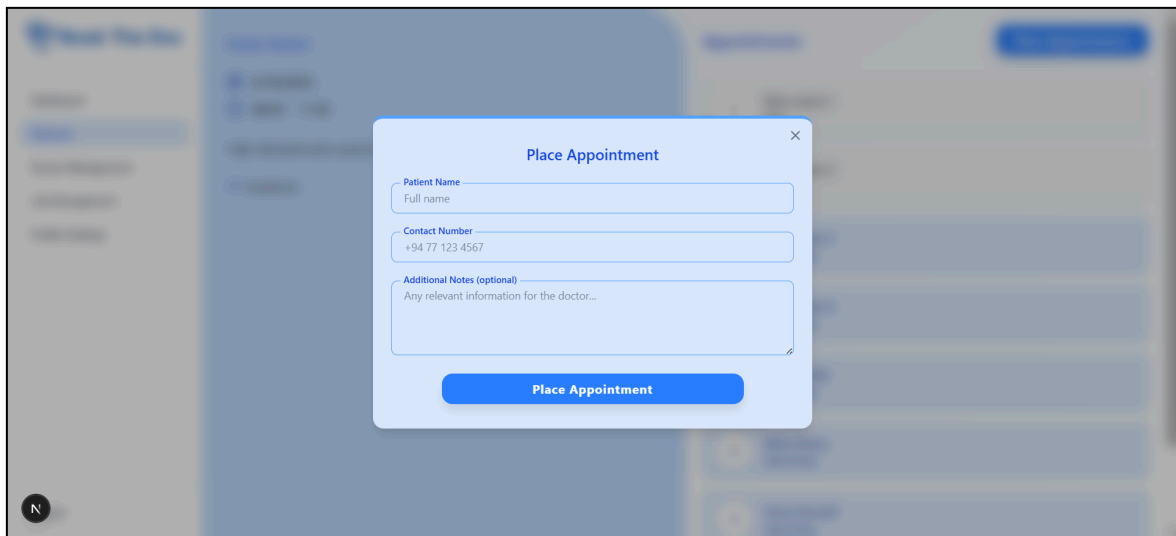


Figure 8 - Place an Appointment Screen

10.2 Database Overview

cluster0.dlmr.mongodb.net

> book-the-doc

Open MongoDB shell

Create collection

Refresh

Sort by

Collection Name

View

centers

Storage size:

Documents:

Avg. document size:

Indexes:

Total index size:

20.48 kB

17

326.00 B

2

73.73 kB

doctors

Storage size:

Documents:

Avg. document size:

Indexes:

Total index size:

20.48 kB

19

226.00 B

2

73.73 kB

doctorsessions

Storage size:

Documents:

Avg. document size:

Indexes:

Total index size:

20.48 kB

5

447.00 B

3

110.59 kB

labsessions

Storage size:

Documents:

Avg. document size:

Indexes:

Total index size:

20.48 kB

5

448.00 B

2

73.73 kB

patientappointments

Storage size:

Documents:

Avg. document size:

Indexes:

Total index size:

20.48 kB

22

284.00 B

2

73.73 kB

Figure 9 - Database Overview

11. Important Implementation Decisions

The development of *Book The Doc* was guided by a series of strategic implementation decisions to ensure scalability, reliability, and user-friendliness. The team selected **Next.js** for the frontend due to its performance and SEO capabilities, and **Node.js** for the backend to support non-blocking I/O and real-time processing, crucial for managing concurrent booking requests.

MongoDB Atlas was chosen for its flexibility in handling semi-structured data and its seamless cloud integration. For secure user authentication, **Google OAuth 2.0** was implemented to allow easy login and protect sensitive medical data. Smart features such as machine learning-based queue time predictions and illness-based clinic recommendations were integrated to enhance user experience. The platform was deployed using **Vercel** and **Railway**, which offered efficient CI/CD pipelines and scalability without high maintenance overhead. These decisions collectively enabled the rapid development and deployment of a robust healthcare appointment system.

12. Timeline

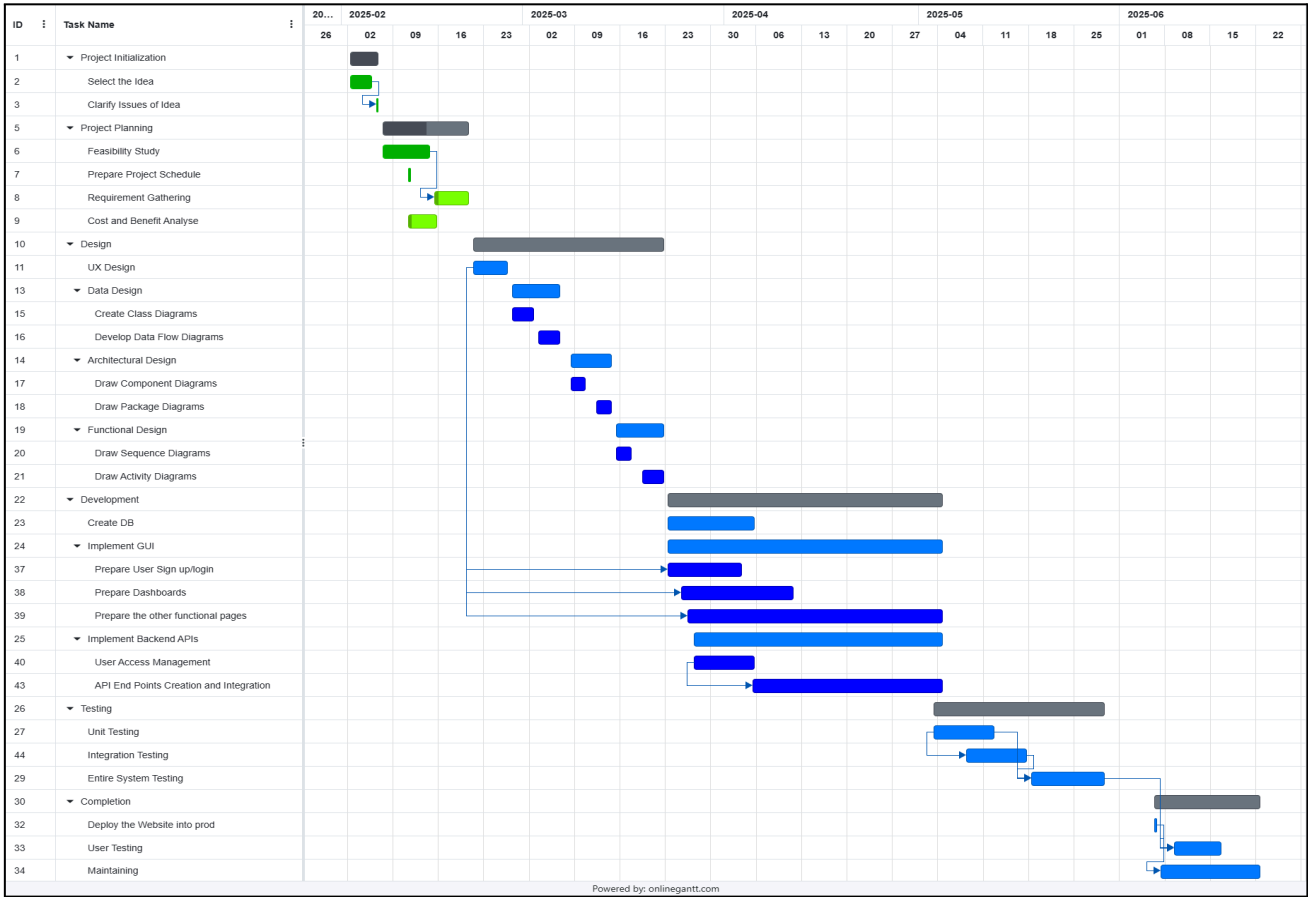


Figure 10 - Grant Chart

13. User Journeys

We maintained a user journey document to maintain Business case requirements and develop them.

	A	B	C	D	E	F	G
	As a User	UJ Number	I want to	Such That	Priority	Authentication Need	Status
5	Patient	UJ001	See Available Medical centers, Doctors and Appointment Schedules	I can see the required booking needed	HIGH	Not Needed	
6	Patient	UJ002	Book an Appointment for a specific Doctor in a specific Medical center Specific Time Slot	I can book an appointment		Needed	
7	Patient	UJ003	Create an account that includes my basic profile and past appointments	To give my details to medical center and to see part appointments	HIGH	Not Needed	
8	Patient	UJ004	See the ongoing session details (Patient Number, Predicted Time) of my Booking	To get an idea on my starting time		Not Needed	
9	Patient	UJ005	see the available medical testings	I can see the required medical testing needed		Not Needed	
10	Patient	UJ006	book a medical testing timeslot	I can book an appointment to test		Needed	
11	Patient	UJ007	edit my profile details	I can change my details when i need it		Needed	
12	Patient	UJ008	See my Predicted time of a particular appointment	i can go to the appointment at the correct time without waiting in a queue		Needed	
13	Medical Center	UJ009	Create a profile and edit it	can show services and appointments provided by the center to user		Not Needed	
14	Medical Center	UJ010	CRUD Doctors	create update delete doctor details when necessary		Needed	
15	Medical Center	UJ011	CRUD doctor session	i can create update delete appointment details when necessary		Needed	
16	Medical Center	UJ012	View the Patients for a booking	to give the approval for the patient when requesting appointment		Needed	
17	Medical Center	UJ013	Edit the Patient Queues for a particular Booking	to show the curent que and more details to user		Needed	
18	Medical Center	UJ014	Update the current session details (Patient Number, Started, Ended)	to keep the current session details updated for the users		Needed	
19	Medical Center	UJ015	CRUD Lab sessions	crud the lab session when center needed		Needed	
20	Patient	UJ016	login to my patient account	get in to personal page and do bookings thoghugh t it			
21	Medical Center	UJ017	login to medical center account	to do center related functionalities			

Figure 11- User Journey Document

14. Profit and Cost Analysis (Business Idea)

Book The Doc is built on a subscription-based business model designed to be financially sustainable while remaining affordable for small and mid-sized medical centers. Clinics and independent doctors can subscribe to monthly or yearly plans, which unlock access to digital appointment scheduling, queue management, and feedback analytics. The use of cloud-based services such as **Vercel**, **Firebase**, and **MongoDB Atlas** keeps operational costs low by leveraging free-tier and pay-as-you-go models. As a result, there are minimal infrastructure and maintenance expenses. The simplicity of the deployment process and reliance on open-source technologies further reduces financial barriers. The platform aims to break even within the first 6 to 12 months by onboarding at least 100 paying clients, with growth potential through the introduction of premium features and tiered plans in the future.

15. Future Enhancements

Several enhancements are planned for future versions of *Book The Doc* to further enrich user experience and system functionality. One key addition will be the integration of **telemedicine** features, allowing video consultations directly through the platform. Another priority is incorporating **Electronic Medical Records (EMR)**, enabling patients and doctors to access health history securely. A built-in **payment gateway** will streamline transactions for consultations and treatments. **Multi-language support** will make the system accessible to a broader user base across Sri Lanka, particularly in rural areas. Finally, the platform will seek integration with **national health APIs** to ensure compliance with healthcare regulations and broaden its utility for public health tracking and coordination.

16. Community Impact

Book The Doc aims to make a tangible impact on Sri Lanka's healthcare system by improving access and reducing inefficiencies. It is especially valuable for rural and underserved communities, where manual appointment systems create barriers to timely care. The platform reduces patient stress by eliminating long queues and simplifying the booking process. It also empowers smaller clinics to digitize their operations, helping them compete with larger hospitals. By promoting the use of digital healthcare tools, Book The Doc encourages broader digital literacy among healthcare providers and patients, contributing to the country's ongoing digital transformation in the medical sector.

17. Challenges Faced

Throughout the development of *Book The Doc*, the team encountered several challenges. One of the most significant was building user trust, particularly among medical staff unfamiliar with digital systems. Demonstrating the system's benefits and ensuring a smooth user experience helped overcome this barrier. Another technical challenge involved managing real-time appointment availability and avoiding double bookings, which was addressed through robust backend logic and database structuring. Tuning the machine learning models for queue prediction required experimentation and adjustment based on test data. Additionally, the team faced time management issues due to academic responsibilities. Weekly reviews and adaptive scheduling were implemented to maintain progress and ensure the timely completion of milestones.

18. Key Takeaway

The development of *Book The Doc* has shown that a focused, tech-enabled approach can effectively modernize healthcare appointment systems in developing countries. A well-planned architecture, combined with user-friendly design and smart features, can address major pain points in the patient journey while empowering healthcare providers. The project highlights the value of simplicity, cost-efficiency, and innovation in solving real-world problems through digital transformation.

19. Results

The team successfully developed and deployed a fully functional prototype, accessible at <https://book-the-doc.vercel.app>. The system supports patient registration, doctor search, appointment booking, real-time queue tracking, and location-based clinic suggestions. Feedback from early users has been overwhelmingly positive, particularly for the platform's intuitive design and smart queue updates. Clinics expressed interest in adopting the platform, validating its relevance and practical utility. These results reflect the project's technical viability and its potential for broader adoption across Sri Lanka.

20. Lessons Learned and Future Directions

Reflecting on the development journey of *Book the Doc*, two key takeaways stood out as particularly valuable for both the current project and future improvements:

- **Value of Early Testing:** Conducting thorough testing during the early stages of development proved essential in identifying bugs and usability issues before they escalated. Establishing a structured testing workflow upfront helped reduce complexity later in the project and ensured a more stable product by deployment.
- **Scalability Considerations from the Start:** It became evident that planning for scalability should be a fundamental part of initial design discussions. Anticipating future growth such as mobile app integration, increased user traffic, and new features allowed us to build a flexible architecture that could evolve without requiring major restructuring.

21. Individual Contribution - Reflective Writing

Guruge S.M.L

During the development of *Book the Doc*, I played a key role in implementing secure user authentication using both Google OAuth and a JWT-based system, which involved managing token lifecycles, secure password handling with bcrypt, and setting up dual-token authentication for improved security. I was responsible for managing backend deployment on Railway, gaining hands-on experience with environment configuration, API endpoint management, and addressing production-level challenges like uptime and request limits. I developed core backend features, including appointment booking logic that ensured proper linking between doctors, patients, and sessions while preventing scheduling conflicts. I also created modular middleware for route protection, token decoding, and role-based access control, which helped make the codebase more scalable and maintainable. Additionally, I implemented the forgot password feature using Nodemailer, enhancing user experience and account recovery security. On the frontend, I used Zustand for clean and efficient state management. I also contributed to setting up Husky pre-push commands and GitHub workflows to enforce coding standards, run checks, and automate deployment steps. This project significantly strengthened my skills in secure and scalable application design, deepened my backend development expertise, and enhanced my ability to collaborate effectively in a team environment..

Sanjana K.Y.C.

In the *Book the Doc* project, my primary focus was on backend API development and database management. I was responsible for structuring the MongoDB database, defining the schemas, and implementing APIs that powered the entire system. This required a strong understanding of RESTful design principles and data modeling.

One of my most significant contributions was creating APIs that handled doctor profiles, session creation, patient data, and booking logic. I ensured that the endpoints were properly validated, error-handled, and aligned with frontend expectations. Managing the database involved careful consideration of data relationships and ensuring referential integrity across collections.

I learned how to write efficient queries, optimize response structures, and maintain clean separation between controller logic and routing. The most challenging part was debugging interdependent API calls and making sure asynchronous logic didn't result in broken flows or race conditions.

Through this project, I strengthened my knowledge of full-stack integration, improved my debugging skills, and became more confident in building reliable backend systems. The

experience emphasized the importance of clear documentation, consistent testing, and collaborative development.

Gallage D

As the primary frontend developer for Book the Doc, I was responsible for designing and developing the user interface and ensuring a smooth user experience across all pages. From the landing page to the appointment screens, I worked on translating wireframes and mockups into fully functional responsive components.

My focus was on ensuring accessibility, responsiveness, and usability. I used technologies like Next.js, Tailwind CSS, and Zustand for state management. I also collaborated closely with the backend team to integrate APIs and display real-time data on the frontend.

One of the most rewarding challenges was ensuring consistency across components and managing different user roles (patients, doctors, centers) in the UI flow. I learned how to optimize performance, use reusable components effectively, and apply modern design principles.

Working on this project gave me hands-on experience in managing frontend complexity, aligning design with functionality, and improving the overall user journey. It also enhanced my team communication skills, especially when resolving integration issues.

Gathsara J.A.S

My contribution to Book the Doc included implementing user profile features (creation and editing), developing doctor detail pages, and handling non-technical project elements such as the slide deck and business case documentation.

From a technical perspective, I ensured that users could manage their profiles seamlessly. This involved both frontend form handling and backend integration, including validations, profile image uploads, and conditional rendering of user data. The doctor detail pages were designed to display qualifications, specialties, and session availability, serving as a key decision-making screen for patients.

On the non-technical side, I played a leading role in preparing the business pitch, designing the presentation, and visualizing user journeys. This helped the team stay focused on real-world use cases and aligned our development tasks with user needs. This project taught me how to balance technical work with design and communication responsibilities. It improved my ability to present ideas clearly and build features that are both functional and user-centered.

Gunathunga W.S.D.

In the Book the Doc project, I was responsible for implementing the core appointment booking logic, as well as handling the creation, updating, and deletion of doctor and session entities (CRUD operations). My goal was to build a seamless flow that allowed patients to book available sessions while enabling centers to manage doctors and their schedules effectively.

Working on the booking system required close collaboration with both frontend and backend teammates. I had to ensure that session timings, availability statuses, and doctor allocations were always synchronized and correctly reflected on the frontend.

Creating doctor and session CRUD APIs helped me solidify my understanding of RESTful practices and taught me how to handle interdependent data structures in MongoDB. I also focused on edge-case handling and error prevention to avoid double-booking or inconsistent session data.

Through this work, I improved my backend logic skills, became more detail-oriented, and learned how to structure features that directly impact user satisfaction. I also gained valuable experience in coordinating across the development lifecycle and resolving integration issues swiftly.

22. Conclusion

Book The Doc is a forward thinking solution to a persistent problem in Sri Lanka's healthcare infrastructure. By combining smart technology with a user-first approach, it delivers a modern, scalable platform that benefits both patients and medical professionals. With its affordable pricing, seamless user experience, and potential for future expansion, Book The Doc positions itself as a transformative tool in healthcare delivery. The successful implementation of this prototype proves that meaningful digital change is achievable and beneficial in even the most traditional sectors.